Case Study

The Clinical and Epidemiological Characteristics of Patients with Vitiligo: A Cross-Sectional Study

Alireza Mohebbipour[1] | Parisa Ahadi[2] | Firouz Amani[3,*] | Aida Adeli[1]| Nafiseh Asadzadeh Maleki[4]

[1]*Department of Medicine, Ardabil Branch, Ardabil Azad University, Ardabil, Iran*
[2]*Department of Pediatrics, School of Medicine, Ardabil University of Medical Sciences, Ardabil, Iran*
[3]*Department of Community Medicine and Biostatistcis, School of Medicine, Ardabil University of Medical Sciences, Ardabil, Iran*
[4]*Department of English Learning, Faculty of Humanities, Male.C., Islamic Azad University, Malekan, Iran*

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ABSTRACT

Vitiligo is the most common depigmenting disorder worldwide, with an estimated prevalence of 0.5% to 2%. Few studies have assessed its epidemiology in Iran, and none specifically in Ardabil Province. This descriptive cross-sectional study aimed to investigate the epidemiological profile of vitiligo patients in Ardabil during the second half of 2022. All patients suspected of vitiligo and referred to a dermatology health center were clinically examined by a dermatologist. Those with a confirmed diagnosis completed a checklist covering variables such as age, sex, occupation, marital status, education, residence, family history, patient complaints, lesion site and pattern, and systemic diseases. Two hundred vitiligo patients participated, including 104 women (52%) and 96 men (48%). The highest frequency was among those over 40 years (28%). Most patients (94%) lived in urban areas, with 43.5% from Ardabil Province. A positive family history was reported in 29% of cases. The most common involvement sites were upper extremities (76.5%), lower extremities (60%), and head/face (58%). Lesion patterns were generalized (75.5%), focal (20%), and segmental (4.5%). Fourteen percent had systemic diseases, primarily hypertension (6%), thyroid disorders (4%), and diabetes (2.5%). The study identified nearly equal gender distribution and an increasing disease frequency with age. Most patients were unemployed or housewives, married, and had education up to diploma level. Urban residence predominated. Family history and generalized lesion patterns were common. Hypertension and thyroid disorders were the most frequent comorbidities among Ardabil vitiligo patients.

Introduction

Vitiligo is the most common depigmenting disorder worldwide [1]. In a study, the prevalence of vitiligo was 0.77% (0.84% in men; 0.67% in women) [2]. The overall prevalence of vitiligo was 3 % (95 %CI: 2 %-4 %) while the prevalence among males and females was 1 % (95 %CI: 1 %-2 %) and 1 % (95 %CI: 1 %-2 %), respectively [3]. Globally, the prevalence is

***Corresponding Author: Firouz Amani**: biostat.f@gmail.com*

estimated between 0.5% and 2%, with peaks of up to 8.8% in India [1,4].

Studies conducted in Mexico and Japan, respectively, indicated high vitiligo rates. A comprehensive inquiry conducted in China revealed an overall prevalence of 0.6% [4]. Most non-segmental vitiligo cases were of sporadic type. Approximately, 15-20% have a family history [5].

Both adults and children, males and females, are equally affected; nevertheless, women are more enthusiastic about getting treatment. Some studies show that the prevalence of vitiligo increases with the age of the patients [1,6].

The primary areas of the body that are affected by this vitiligo include the face, joints, hands, and feet. Virtually, the superficial dermis of parts of the body composed of mucosa may be involved [7].

Vitiligo is classified by depigmentation patterns into focal, segmental, generalized, and universal forms. It is not considered as a physical disability but culminates in momentous psychosocial challenges. Various diseases like hyperthyroidism, hypothyroidism, diabetes, Addison's disease, pernicious anemia, alopecia areata, Hodgkin's lymphoma, multiple myeloma, and thymoma are associated with vitiligo, supporting the autoimmune hypothesis [8].

Vitiligo results from loss of functional melanocytes, possibly due to metabolic abnormalities, oxidative stress, inflammatory mediators, cell detachment, and autoimmune responses. The contribution of each mechanism is debated [9].

Oxidative stress in melanocytes triggers local inflammation and innate immune activation, which, in genetically susceptible individuals, leads to melanocyte-specific cytotoxic immune responses [10].

Progressive melanocyte loss may occur via immune attack or cell degeneration and detachment. Segmental and non-segmental vitiligo may have distinct pathogenic mechanisms; segmental vitiligo possibly involves neuronal or somatic mosaicism [11].

Recent data show overlapping inflammatory pathogenesis involving the early release of pro-inflammatory cytokines and neuropeptides, vascular dilation, and a subsequent immune response [12].

The previous literature validated that there are virtually few epidemiological studies on vitiligo in Iranian population in general, and none in Ardabil in particular. This study attempts to shed light on the examination of the epidemiological status of vitiligo in Ardabil Province.

Methods

This descriptive cross-sectional study was comprised of all vitiligo patients attending the dermatology clinic during the second half of 2022. Based on the estimated national prevalence of vitiligo at 0.5% to 1%, and the sample size formula by Cochrane, the calculated sample size was 200. The sampling procedure had continued until the researchers could reach to the specified number over six months. Data collection tools included pre-prepared checklists. The clinical diagnosis of vitiligo was performed by a dermatologist. The determined variables were age, sex, occupation, marital status, education, residence, family history, patient complaints, lesion site, lesion pattern, and the presence of systemic disease. The obtained data were analyzed using SPSS version 25 for frequencies, percentages, means, graphs, and tables.

Results

Among all patients, 104 were women (52%) and 96 were men (48%), showing a slightly higher female prevalence; however, the difference between two genders was not significant (Figure 1). Regarding the age of the patients, the highest prevalence was those who were more than 40 years old (56 patients, 28%) and the lowest was among patients who were less than 10 years old (17 patients, 8.5%). In terms of the patients' occupations and marital status, housewives (59 patients, 29.2%), unemployed (60 patients, 30%), and married (116 patients, 58%) represented the highest ranges. Most of the patients had diploma or below (104 patients, 51.5%). Considering the residence status of the patients, 188 of them lived in urban areas

(94%) and 87 of them (43.5%) were from Ardabil city. A family history was observed in 58 patients (29%). Considering the population size of Ardabil Province, approximately 1,285,000 people, the estimated prevalence rate of vitiligo was 0.02% or about 15.6 per 100,000. Based on Table 1, most initial complaints were related to upper extremities (54%). As indicated in Table 2, the highest amount of involvement at the time of study was in all parts of the patients' bodies (37%). Table 3 shows that the majority of lesion patterns (75.5%) had generalized vitiligo. As a final point, as indicated in Table 4, among those with systemic diseases (14% out of 100 %), the hypertension disease was the most common type (6%).

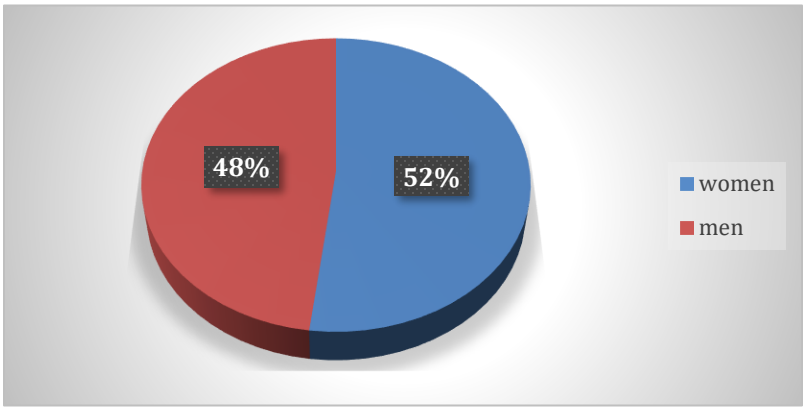


Figure 1. The gender distribuation of stuided patients

Table 1 Frequency distribution of initial complaint site

Initial Complaint Site	n	%
Lower extremities	38	19
Upper extremities	108	54
Head and face	54	27
Total	200	100

Table 2 Frequency distribution of current lesion sites

Lesion Site	n	%
Lower extremities	5	2.5
Upper extremities	38	19
Head and face	42	21
Lower and upper extremities	41	20.5
Whole body	74	37
Total	200	100

Table 3 Frequency distribution of lesion patterns

Pattern	n	%
Focal	40	20
Segmental	9	4.5
Generalized	151	75.5
Total	200	100

Table 4 Frequency distribution of systemic diseases among patients

Disease	n	%
Hypertension	12	6
Diabetes	5	2.5
Dyslipidemia	2	1
Anemia	1	0.5
Thyroid disease	8	4
No systemic disease	172	86
Total	200	100

Discussion

In the present study, the ratio of patients with vitiligo between females and males was nearly equal, with 52% females and 48% males. This finding is consistent with the study conducted by Lee *et al.* in South Korea, who also observed no difference between genders in the prevalence of vitiligo [13]. However, findings from other studies concerning the gender distribution of vitiligo patients are conflicting; for instance, Fahaad *et al.* [14] in Saudi Arabia and Wang *et al.* [15] in China, revealed that males were more affected than females, while other studies, such as those conducted by Nagaty *et al.* [16] in Egypt, Rahman *et al.* [17] in Bangladesh, and Singh *et al.* [18] in India, do not align with the results of this study, reporting a higher prevalence of vitiligo among females [13-18].

Two recent studies conducted in Iran by Shaban *et al.* [8] in Birjand and Shamseddini *et al.* [19] in Gorgan, likewise, reported dissimilar conclusions compared to the present study, indicating that the majority of vitiligo patients were female in these two studies. Overall, most reference sources' data are in accordance with the results of this study and state that both males and females are equally affected by this disease. The slightly higher proportion of females in the present study may be due to women paying more attention to their appearance and consequently visiting doctors more frequently [19,20].

The findings of the present study demonstrate an increasing trend in vitiligo prevalence with age, confirming that the lowest prevalence was observed in the age group below 10 years old (8.5%). With the increase in the age of the patients, vitiligo prevalence gradually and steadily enhanced, reaching its highest frequency in the age group above 40 years old (28%). The results were consistent with the those of Wang *et al.* [15] in China, demonstrating a high range of vitiligo prevalence in older patients. However, the most affected age group in Fahaad *et al.* [14] in Saudi Arabia was 31–40 years (28.7%), and in Rahman *et al.* [17] in Bangladesh was 11–20 years (44%), which were not in accordance with this study's findings. The results were not in line with the study in South Korea, which showed a bimodal (two-peak) age distribution of vitiligo. Furthermore, the highest proportion of vitiligo patients in Birjand was in the 20–29 years group (39.7%), and in Gorgan under 20 years (72%), both of which did not agree with the present study's findings. The increase in vitiligo prevalence concerning the age of the patients in the present study might be due to a cumulative effect, since vitiligo is usually a chronic and often lifelong condition [14,15,17,19,20]. Occupationally, the most common categories among vitiligo patients in this study were unemployed (30%), housewives (29.2%), and self-employed (26%). Similar to the findings of the present study, housewifery is a common occupation among vitiligo patients in studies conducted in Egypt and Saudi Arabia. However, in a study in Bangladesh, students were the predominant group affected by vitiligo. Additionally, 14.5% of our patients were employees; this percentage was reported as 30.4% in Egypt, which is higher than this study's data. Consistent with the results of the present study, the majority of vitiligo patients lived in urban areas (94%). This finding was in consistent with Singh *et al.*'s (2011) study in India, where most patients also came from

urban regions. Furthermore, the current study showed that the frequency of vitiligo patients varied among geographical areas, with most patients from Ardabil Province (43.5%) and Parsabad (12.5%), while the frequency amount obtained from other provinces was less than or equal to 5%. Previous studies conducted in China also demonstrated differences among regions and cities regarding vitiligo prevalence. However, precise conclusions about geographic distribution require population-based studies. In the present study, most patients (54%) reported that the initial site of disease onset was the upper extremities, followed by the head and face (27%) and lower extremities (19%). In contrast to our findings, Singh *et al.* [18] in India reported that the lower extremities were the most common initial site in 44% of patients. The most common sites of vitiligo involvement in the present study were the upper extremities (76.5%), lower extremities (60%), and head and face (58%), which aligned with the study conducted by Fahaad *et al.* (2015) in Saudi Arabia, where the most common sites were the upper extremities (68.4%), lower extremities (58.5%), and head and face (55.6%) . However, a study on vitiligo patients in Gorgan reported a different order: head and neck, upper extremities, lower extremities, and finally the trunk [13-18]. The results of the present study showed that generally the head and face of 58% of examined vitiligo patients were afflicted by this depigmenting disorder (21% of head and face, and the remaining 37% of other body areas). The head and face are the most commonly affected parts of vitiligo patients, which have a momentous impact on the quality of life and psychological well-being of patients. Most studies have confirmed this issue; for instance, the amount of involvement of the head and face was 54% in the investigation conducted in Birjand, 53.3% in Egypt , and 45.6% in Saudi Arabia. In the present study, 79% of patients had involvement of distal limbs (upper and lower extremities). The rate of distal limb involvement was higher in the study conducted in Birjand (83%), while it was lower in studies conducted in Saudi Arabia (63.4%) and Egypt (60%). The higher prevalence of vitiligo lesions in distal limbs, particularly upper limbs and head and face, suggests that environmental factors such as exposure to ultraviolet radiation may play a significant role in the etiology of vitiligo. The involvement pattern in 75.5% of patients in the present study was generalized. The prevalence of generalized vitiligo in Birjand was reported as 73%, similar to this study's findings; however, Nagaty *et al.* (2020) in Egypt reported a higher prevalence of 80%. Furthermore, the rates of generalized vitiligo in studies conducted in Gorgan (49.1%) and China (35.2%) were both lower than the current study's findings [8,15-16,19].

Focal vitiligo was noticeable in 20% of patients in this study, whereas the frequency of this type in Wang *et al.* [13] in China was higher (36.1%), and lower in an Egyptian study (6.7%) [15]. Segmental vitiligo was observed in 4.5% of patients in this study. The frequency of segmental vitiligo showed a discrepancy in other studies ranging from low values such as 1% in Saudi Arabia and 2.5% in China to higher values of 6.7% in Egypt and 14.5% in India. In a study conducted in Birjand, the frequency of segmental vitiligo was reported as 6%, closer to this study's findings in comparison to other study outcomes [15-19]. A positive family history of vitiligo was recognizable in 29% of patients in this study. Frequencies of positive family history in Saudi Arabia, China, Egypt, India, Birjand, and Gorgan were lower at 5.9%, 8.9%, 14.3% , 22% , 23.6% , and 24.8% respectively. However, a study conducted in Bangladesh reported 32% positive family history which was in line with the results of this study. Overall, having a positive family history of vitiligo is recognized as a risk factor for the disease; studies have shown that the risk of developing vitiligo for first-degree relatives of a patient is increased tenfold compared to unrelated individuals with comparable genetic backgrounds [13-18,20,21].

In the present study, 14% of vitiligo patients had concurrent systemic diseases, most commonly hypertension, thyroid diseases, diabetes, dyslipidemia, and anemia. This finding is supported and confirmed by multiple studies; for example, a population-based study in South Korea found that vitiligo is associated with various autoimmune and non-

autoimmune diseases such as thyroiditis, diabetes, hypertension, anemia, as well as atopic dermatitis and psoriasis. In a study conducted by Fahaad *et al.* (2015) in Saudi Arabia, 17.8% of vitiligo patients had other comorbid diseases, which were mainly diabetes, anemia, hypertension, and hypothyroidism, aligning with our findings [14].

Singh *et al.* in India reported additional comorbidities such as rheumatoid arthritis, and noted a higher prevalence of thyroid diseases (12% in their study versus 4% in the present study) [18].

The crucial strong point of this study is that it provides the epidemiological profile of vitiligo patients in the Ardabil region for the first time. Like any other study, the current research was also affected by some limitations such as the short duration time of the study and the small size of the population. To achieve to more comprehensible and generalizable results, it is highly recommended to conduct this study over an extended period of time with a larger sample size of patients.

Conclusion

This study identified several epidemiological aspects of vitiligo in Ardabil Province. Gender distribution was nearly equal; frequency increased with age. Most patients were unemployed or housewives, married, had education up to diploma, and lived in urban Ardabil. About 30% had positive family history. The upper extremities were the most common onset and involvement sites. Generalized vitiligo was most frequent, segmental cases accounting for about 5%. Hypertension and thyroid disorders were the common comorbidities. This is a minor study with a small sample size in Ardabil and to obtain a comprehensive profile of vitiligo patients, multicenter studies with larger sample sizes in Ardabil and other provinces are suggested. Likewise, designing studies to assess the effects of vitiligo on psychological health and quality of life is indispensable.

ORCID

Alireza Mohebbipour
https://orcid.org/0009-0009-0692-4435
Parisa Ahadi
https://orcid.org/0000-0001-9651-9349
Firouz Amani
https://orcid.org/0000-0002-0989-1052
Nafiseh Asadzadeh Maleki
https://orcid.org/0009-0002-9561-9879

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